

Biology Midterm — Student WD_S02

Cell membranes are dynamic phospholipid bilayers that exhibit selective permeability, meaning they carefully regulate what substances move into and out of the cytoplasm. Uncharged, small molecules like O₂ and CO₂ diffuse across without assistance, whereas larger or charged compounds rely on facilitated diffusion or ATP-powered active transport mechanisms. Proteins embedded in the membrane act as signal receptors, initiating transduction cascades, and cholesterol molecules interspersed in the bilayer regulate its fluidity and mechanical stability.